

6.8 Splitting and joining strings ^

Settings

Beta features

split()

The string method `split()` splits a string into a list of tokens.

token

Each token is a substring that forms a part of a larger string.

separator

A separator is a character or sequence of characters that indicates where to split a string into tokens.

PARTICIPATION ACTIVITY

6.8.1: Splitting a string into tokens.

Start

2x speed

```
string = "Music/artist/song.mp3"
my_tokens = string.split("/")
```

```
string = "I love python"
my_tokens = string.split()
```

"Music/artist/song.mp3"

"I love python"

my_tokens = ["Music", "artist", "song.mp3"]

my_tokens = ["I", "love"]

Captions ^

1. Original string contains a pathname to an mp3 of your favorite song.
2. The pathname is split using the delimiter " / ".
3. The variable my_tokens is assigned with the 3 tokens as a list of strings.
4. When split() is called with no argument, the delimiter defaults to a space character.

Figure 6.8.1: String split example.

```
url = input("Enter URL:\n")

tokens = url.split("/") # Uses "/" separator
print(tokens)
```

```
Enter URL: http://en.wikipedia.org/
["http:", "", "en.wikipedia.org/"]
...
Enter URL: en.wikipedia.org/wiki/
["en.wikipedia.org", "wiki", ""]
```

Try 6.8.1: More string splitting.

Run the following program and observe the output. Edit the program by changing the split() method and observe the output.



▶ Run

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join()

The `join()` string method performs the inverse operation of `split()` by taking an array of strings and joining them together to create a single string.

**PARTICIPATION
ACTIVITY**

6.8.3: String join() method.

Start

2x speed

```
web_path = [ "www.website.com", "profile", "settings" ]
separator = "/"
url = separator.join(web_path)
```

```
url = "www.website.com/profile/settings"
```

Captions ^

1. web_path is a list of strings that form the path of the web page.
2. Create a string with the separator "/".
3. Then join() concatenates the list of strings with the separator "/".

Figure 6.8.2: String join() example: Comparing join vs. loops.

The following programs are equivalent, but join() is a simpler approach that uses less code and is

```
phrases = ["To be, ", "or not to be.\n", "That is the question."]

sentence = ""
for phrase in phrases:
    sentence += phrase
print(sentence)
```

```
phrases = ["To be, ", "or not to be.\n", "That is the question."]

sentence = "".join(phrases)
print(sentence)
```

Figure 6.8.3: Splitting and joining: Replacing separators.

```
path = input("Enter file name: ")
new_separator = input("Enter new separator: ")
tokens = path.split("/")
print(new_separator.join(tokens))
```

```
Enter file name: C:/Users/Wolfman/Documents/report.
Enter new separator: \
C:\\Users\\Wolfman\\Documents\\report.pdf
```

Figure 6.8.4: Splitting and joining: Editing tokens.

```
url = input("Enter Wikipedia URL: ")
tokens = url.split("/")
if "wiki" != tokens[3]:
    tokens.insert(3, "wiki")
    new_url = "/".join(tokens)
    print(f"{url} is not a valid address.")
    print(f"Redirecting to {new_url}")
else:
    print(f"Loading {url}")
```

```
Enter Wikipedia URL: http://en.wikipedia.org/wiki/R
Loading http://en.wikipedia.org/wiki/Rome
...
Enter Wikipedia URL: http://en.wikipedia.org/Rome
http://en.wikipedia.org/Rome is not a valid address
Redirecting to http://en.wikipedia.org/wiki/Rome
```

**CHALLENGE
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6.8.1: String split and join.

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Start

Type the program's output

```
item_info = "Headphones 9 19"

item_tokens = item_info.split()
item = item_tokens[0]
quantity = item_tokens[1]
price = item_tokens[2]

print(f"{item} stock: {quantity}")
print(f"Price: {price}")
```

```
Headphones stock: 9
Price: 19
```

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ACTIVITY**

6.8.2: Splitting and joining strings.

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Start

String `key_value_pair` is read from input. Split `key_value_pair` into tokens using `:` as the separator. The result is:

Ex: If the input is `Year of birth: 1972`, then the output is:

```
['Year of birth', '1972']
```

```
1 key_value_pair = input()
2
3 """ Your code goes here """
4
5 print(tokens_list)
```

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